

Accelerated Quantum Supercomputing in Action: A Hands-On Tutorial on Scalable Hybrid Workflows

Monica VanDieren¹ and Pooja Rao¹
¹*NVIDIA, Santa Clara, CA*

Pranav Gokhale²
²*Inflection, Chicago, IL*

Ermal Rrapaj³
³*NERSC, Lawrence Berkeley National Lab, Berkeley, CA*

I. Detailed Outline

[Link to the official repo for the tutorial.](#) The repo will be made public close to the tutorial.

Date: Sun Nov 16, 1:30 - 5pm

• Hour 1 : Accelerated Quantum Supercomputing (NV, NERSC)

- ▷ Onboarding with NERSC
 - Logging in, running interactive sessions, submitting jobs, and post-processing
 - Using Jupyter notebooks on Perlmutter using a previously tested container
- ▷ Introduction to AQSC
 - Overview
 - Emphasis on hybrid workflows
 - Mention of QEC efforts with Inflection
- ▷ Hands-on with CUDA-Q
 - Introductory Jupyter notebooks with simple hybrid workflows such as VQE
 - Different backends for executing quantum circuits
 - CUDA-QX intro
 - Scaling simulations from 1 to 100s of GPUs across nodes with minimal changes in workflow.

• Hour 2: AI for Scalable Quantum Algorithms (NV, NERSC)

- ▷ Hybrid algorithms
 - Introduce GQE
 - Compare and contrast with VQE
- ▷ GQE hands-on
 - Algorithm design
 - Performance
 - Parallelization opportunities
- ▷ Advanced
 - Quickly highlight AFQMC, QAOA-GPT, etc.
 - Point to more advanced material

• Hour 3: Logical Qubit Experiments (Inflection, NERSC)

- ▷ 1. Running a logical qubit VQE ground state prep with `cudaq`:
 - Materials science VQE application (single-impurity Anderson model) in CUDA-Q using two logical qubits. The demo is contained in a Jupyter notebook for generating and noisily simulating logically encoded circuits with the `[[4, 2, 2]]` code. Users can use a GPU to leverage the CUDA-Q GPU backend for the Hamiltonian Variational Ansatz optimization in `cudaq-solvers`.

- ▷ 2. Running decoding for the many-hypercube QEC code using `cudaq-qec` and Infleqtion’s `qLDPC` with NERSC’s Perlmutter Supercomputer:
 - Users will be able to execute a Python script to run decoding of real data from Infleqtion’s Sqale QPU. In particular, the script will decode the $[[16, 4, 4]]$ many-hypercube logical state preparation reported in <https://arxiv.org/abs/2509.13247>. It uses `cudaq-qec`’s GPU-accelerated belief propagation (BP) decoder, along with the `qLDPC` library, to demonstrate a pipeline for decoding syndromes and retrieving logical state outcomes. We will show how to set up the environment for running the decoder on a GPU on Perlmutter in interactive mode.
- ▷ 3. Running an atom rearrangement experiment on Infleqtion’s Sqale QPU:
 - A Jupyter notebook will showcase the atom rearrangement capability of Infleqtion’s Sqale QPU, a feature leveraged in executing Infleqtion’s logical qubit experiments. Users will be able to submit a 2D bitmap array job encoding a specified pattern via the presence or lack of an atom in the Sqale QPU array. To do so, users will be able to acquire a Superstaq API key to gain cloud access to the QPU. Instructions to do so can be found [here](#). Atom array capture results will be sent to the email associated with the user’s API key, with lead time subject to the QPU’s queue.